

2017-2018 – DEEQA – TSE – Advanced Macroeconomics

Lecture 5 : Dynamics with Interactions

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0. Introduction

- ▶ Here I present first abstract static and dynamic models with strategic interactions
- ▶ These models have a macro flavour : a large number of atomistic agents
- ▶ I discuss the role of complementarities in generating multiplicity, multipliers, rich dynamics
- ▶ I then present an extended NK model that features rich dynamics (up to limit cycles)
- ▶ I show that it can produce boom-bust cycles and peaks in spectral density.

1. Static model - COOPER & JOHN, 1988

Setup

- ▶ N players with payoffs function $V(e_j, e)$, where $e = \frac{1}{N} \sum_{j=1}^N e_j$
- ▶ $e \in [0, E]$
- ▶ $V_{11}(e_j, e) < 0$
- ▶ $e_j^*(e)$ is the best response function, implicitly defined from

$$V_1(e_j, e) = 0;$$

1. Static model - COOPER & JOHN, 1988

Symmetric Cooperative Equilibrium

- ▶ SCE is simply the planner solution

$$\max_e NV(e, e)$$

- ▶ FOC is

$$V_1(e, e) + V_2(e, e) = 0$$

- ▶ SOC is

$$V_{11}(e, e) + 2V_{12}(e, e) + V_{22}(e, e) < 0$$

- ▶ Assume for existence of an interior SCE

$$\lim_{e \rightarrow 0} V_1(e, e) + \lim_{e \rightarrow 0} V_2(e, e) > 0$$

$$\lim_{e \rightarrow E} V_1(e, e) + \lim_{e \rightarrow E} V_2(e, e) < 0$$

1. Static model - COOPER & JOHN, 1988

Some definitions

- ▶ *Positive spillovers* : $V_2(e_j, e) > 0$
- ▶ *Negative spillovers* : $V_2(e_j, e) < 0$
- ▶ *Strategic complementarity* : $V_{12}(e_j, e) > 0$
- ▶ *Strategic substitutability* : $V_{12}(e_j, e) < 0$
- ▶ *Multiplier* :
 - × $\frac{\partial e_j^*(e; \theta)}{\partial \theta} = \frac{V_{1\theta}}{V_{11}}$ is the partial equilibrium response of e to θ
 - × $\frac{\partial e}{\partial \theta} = \frac{V_{1\theta}}{V_{11} + V_{12}}$ is the general equilibrium response of e to θ
 - × There are multipliers when the general equilibrium response is in the same direction as the partial equilibrium one, and greater.
 - × An example of no multipliers would be the consumption of apples when taste for apple increases in a walrasian equilibrium.

1. Static model - COOPER & JOHN, 1988

Individual decision

- ▶ Agent aims at $\max_{e_j} V(e_j, e)$
- ▶ Agent j takes optimal action *given* the actions of others.
- ▶ Either she takes into account her effect on e or she does not
 - × If she does, FOC is

$$V_1(e_j, e) + V_2(e_j, e) \frac{\partial e}{\partial e_j} = V_1(e_j, e) + \frac{1}{N} V_2(e_j, e) = 0$$

- × If she does not

$$V_1(e_j, e) = 0$$

- × I focus on the second case (simple, but does not affect results)
- × N large $\rightsquigarrow$ close to the second case, which is the maximum departure from social optimum, as we will see

1. Static model - COOPER & JOHN, 1988

Best response

- ▶ Given by $V_1(e_j, e) = 0$
- ▶ Slope is (by full differentiation)

$$\rho = \frac{de_j}{de} = -\frac{V_{12}}{V_{11}}$$

- ▶ Best response can be increasing, decreasing, non monotonous, non linear,...

1. Static model - COOPER & JOHN, 1988

Symmetric Nash Equilibrium

- ▶ SNE defined by

$$V_1(e, e) = 0$$

- ▶ to be compared to the SCE

$$V_1(e, e) + V_2(e, e) = 0$$

1. Static model - COOPER & JOHN, 1988

Results

- ▶ Prop 1 : if there are spillovers, then the SNE is inefficient
- ▶ Prop 2 : If spillovers are positive, then the SNE levels of e are too low
- ▶ Prop 3 : Strategic complementarities are necessary and sufficient for multipliers
- ▶ Note : When there are positive spillovers and strategic compl., may want to refer to such equilibrium outcome as exhibiting “*deficient demand*”

1. Static model - COOPER & JOHN, 1988

Results

- ▶ Prop 4 : There can be multiple equilibria. Strategic complementarities (and $-\frac{V_{12}}{V_{11}} > 1$ at some point) are necessary for multiplicity
- ▶ Prop 5 : If positive spillovers and multiple equilibria, then equilibria are Pareto ranked with higher actions being better.
- ▶ Corollary : If $e_j^*(e) = e$ over an interval, and there are positive spillovers, then there is a continuum of equilibrium increasing in e .

1. Static model - COOPER & JOHN, 1988

Relation the Walrasian Equilibrium

- ▶ A Walrasian equilibrium can be seen as a special case of the above game situation.
- ▶ In this case, we can see agents as choosing actions – expenditures or supply decisions – taking prices as given, but where prices depend on aggregate actions
- ▶ What differentiates the Walrasian equilibrium is at an equilibrium $V_2 = 0$, ie, there are no spillovers.
- ▶ Therefore, the Walrasian equilibrium is efficient
- ▶ This is not because agents are small, but because the effects of ones actions - at the margin - does not have a positive nor negative effect on the others.

1. Static model - COOPER & JOHN, 1988

Relation the Walrasian Equilibrium

- ▶ Here is a fully specified example.
- ▶ Consider a simple consumption labor supply problem, where agent j has preferences $U(c_j, e_j)$, and has the budget constraint

$$c_j = we_j + \frac{\Pi}{N}$$

where $\frac{\Pi}{N}$ is the share of profits

- ▶ There are N agents and they are all the same
- ▶ We will focus on symmetric allocations $e_j = e \forall j$
- ▶ There is a competitive firm with technology $F(Ne)$
- ▶ Firm profit maximisation leads to

$$F'(Ne) = w \text{ or equivalently } w = w(Ne)$$

- ▶ The consumption good is the numéraire

1. Static model - COOPER & JOHN, 1988

Relation the Walrasian Equilibrium

- ▶ In this setup, the V function is

$$V(e_j, e) = U \left(w(Ne)e_j + \frac{\Pi(N, e)}{I}, e_j \right)$$

- ▶ The individual FOC is

$$wU_1 + U_2 = 0$$

1. Static model - COOPER & JOHN, 1988

Relation the Walrasian Equilibrium

- ▶ Let us now consider the social optimum
- ▶ We have $V(e, e) = U\left(w(Ne)e + \frac{\Pi(N, e)}{I}, e\right)$
- ▶ The Planner FOC is

$$wU_1 + U_2 + \sum_{j=1}^N U_1 \left[\frac{\partial w}{\partial e} e + \frac{(F'(Ne) - w) \frac{\partial Ne}{\partial e}}{N} - \frac{\partial w}{\partial e} \frac{Ne}{N} \right] = 0$$

- ▶ Because $F'(Ne) = w$, this is

$$wU_1 + U_2 + \sum_{j=1}^N U_1 \left[\frac{\partial w}{\partial e} e - \frac{\partial w}{\partial e} \frac{Ne}{N} \right] = 0$$

which rewrites

$$wU_1 + U_2 + \sum_{j=1}^N U_1 \left[\frac{\partial w}{\partial e} (e - e) \right] = 0$$

1. Static model - COOPER & JOHN, 1988

Relation the Walrasian Equilibrium



$$wU_1 + U_2 + \underbrace{\sum_{j=1}^N U_1 \left[\frac{\partial w}{\partial e}(e - e) \right]}_0 = 0$$

- ▶ So that we are back to the competitive equilibrium condition.
- ▶ As we see, the actions of the others have an impact on one's payoff V , but nevertheless there are no inefficiencies at the Walrasian equilibrium.

1. Static model - COOPER & JOHN, 1988

Relation to Global Games

- ▶ This setup extends naturally to situation with imperfect information.
- ▶ For example, the θ in $V(e_j, e; \theta)$ may be a latent variable, which is only observed with error by agents.
- ▶ This is the global games framework.
- ▶ Some of the insights from the global games setup
 - × Imperfect information may eliminate multiple equilibria.
 - × Imperfect information affects the strength of complementarities (now reaction is a function of the expectation of the others actions, which depends on one's expectations of others expectation). This can cause expectation to play the role of a state variable
 - × This may allow for shocks to higher order beliefs (an example is ANGELETOS & LA'O (2013))

2. A framework for Dynamics with Strategic Interactions

Environment

- ▶ N players
- ▶ Each agent accumulates E_i by playing e_i
- ▶ Decision rule and law of motion for E are

$$E_{it+1} = (1 - \delta)E_{it} + e_{it} \quad (1)$$

$$e_{it} = \alpha_0 - \alpha_1 E_{it} + \alpha_2 e_{it-1} + F(e_t) \quad (2)$$

with $e_t = \frac{\sum e_{jt}}{N}$

- ▶ all parameters are positive, $\delta < 1$, $\alpha_1 < 1$, $\alpha_2 < 1$.
- ▶ $\alpha_1 E_{it}$: Optimal size argument
- ▶ $\alpha_2 e_{it-1}$: Adjustment cost argument
- ▶ $F(e_t)$: Strategic Interactions
 - × $F' > 0$: strategic complementarities
 - × $F' < 0$: strategic substitutabilities

2. A framework for Dynamics with Strategic Interactions

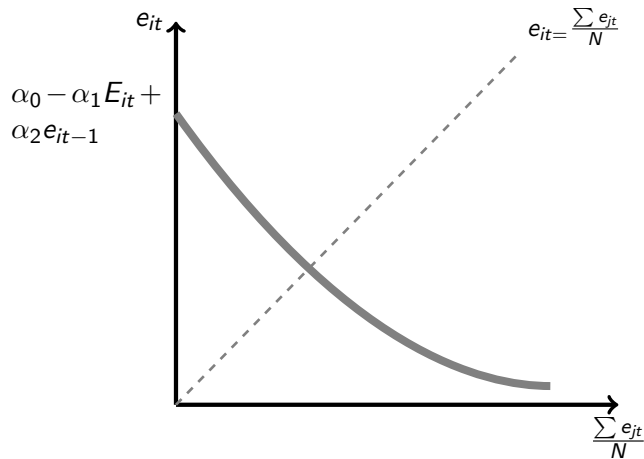
Environment

$$\begin{aligned}E_{it+1} &= (1 - \delta)E_{it} + e_{it} \\ e_{it} &= \alpha_0 - \alpha_1 E_{it} + \alpha_2 e_{it-1} + F(e_t)\end{aligned}$$

- ▶ Define (E^s, e^s) as the steady state of the linear model where $F(\cdot) = 0$
- ▶ Normalize $F(e^s) = 0$

2. A framework for Dynamics with Strategic Interactions

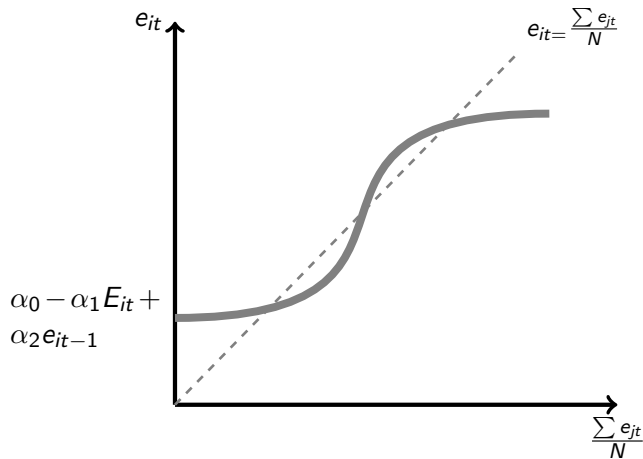
"Best response" rule for a given history - Strategic substitutability



$$e_{it} = \alpha_0 - \alpha_1 E_{it} + \alpha_2 e_{it-1} + F\left(\frac{\sum e_{jt}}{N}\right)$$

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"Best response" rule for a given history - Multiple Equilibria

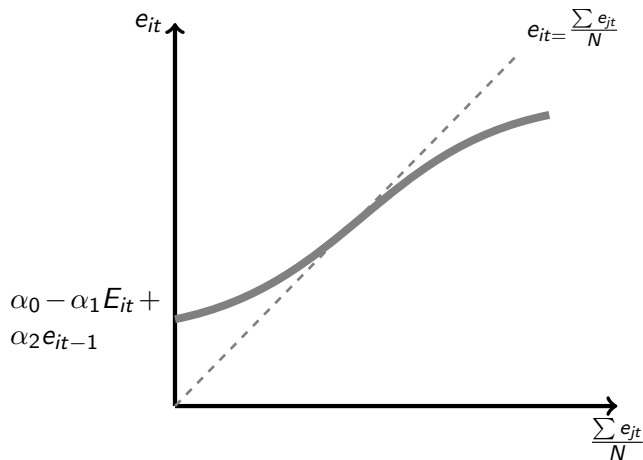


$$e_{it} = \alpha_0 - \alpha_1 E_{it} + \alpha_2 e_{it-1} + F\left(\frac{\sum e_{jt}}{N}\right)$$

2. A framework for Dynamics with Strategic Interactions

"Best response" rule for a given history - Strategic complementarity

- We restrict to the case where $F'(\cdot) < 1$, so that there are never multiple equilibria *within* period t



2. A framework for Dynamics with Strategic Interactions

One-Dimension Case

- ▶ Assume $\alpha_2 = 0$
- ▶ Then the model is

$$\begin{aligned}E_{it+1} &= (1 - \delta)E_{it} + e_{it} \\e_{it} &= \alpha_0 - \alpha_1 E_{it} + F(e_t)\end{aligned}$$

- ▶ It boils down to one order one dynamic equation in E_{it}

$$E_{it+1} = \alpha_0 + (1 - \delta - \alpha_1)E_{it} + F(E_{t+1} - (1 - \delta)E_t)$$

- ▶ So that, for symmetrical allocations

$$E_{t+1} = \alpha_0 + (1 - \delta - \alpha_1)E_t + F(E_{t+1} - (1 - \delta)E_t)$$

2. A framework for Dynamics with Strategic Interactions

One-Dimension Case

- Assume no strategic interactions : $F(\cdot) \equiv 0$

$$E_{t+1} = \alpha_0 + (1 - \delta - \alpha_1)E_t$$

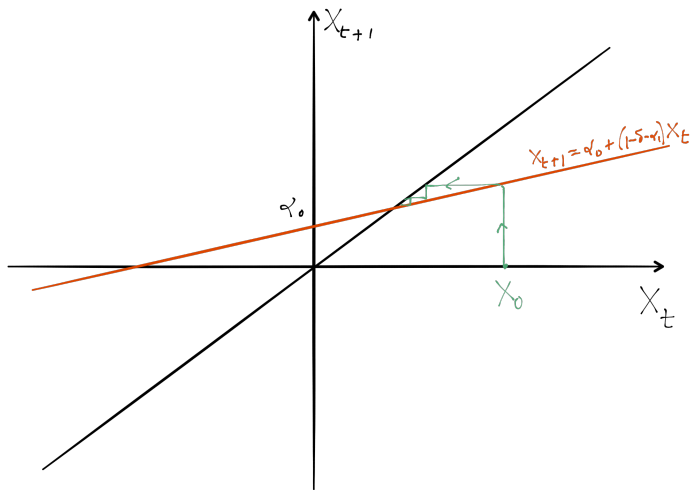
- Note that $(1 - \delta - \alpha_1)$ can be either > 0 or < 0 .

Proposition 1

The Economy is globally stable as $|1 - \delta - \alpha_1| < 1$

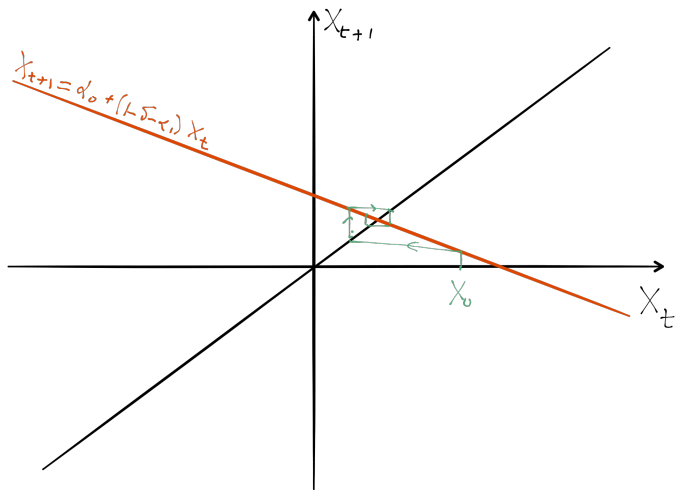
2. A framework for Dynamics with Strategic Interactions

One-Dimension Case



2. A framework for Dynamics with Strategic Interactions

One-Dimension Case



2. A framework for Dynamics with Strategic Interactions

One-Dimension Case

- Reintroduce strategic interactions

$$E_{t+1} = \alpha_0 + (1 - \delta - \alpha_1)E_t + F(E_{t+1} - (1 - \delta)E_t)$$

- Linearize around the steady state E^s

$$E_{t+1} = (\alpha_0 + \delta E^s) + (1 - \delta - \alpha_1)E_t + F'(E^s)(E_{t+1} - (1 - \delta)E_t)$$

- so that

$$E_{t+1} = \frac{\alpha_0 + \delta E^s}{1 - F'(E^s)} + \frac{((1 - \delta)(1 - F'(E^s)) - \alpha_1)}{1 - F'(E^s)} E_t$$

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One-Dimension Case

$$E_{t+1} = \frac{\alpha_0 + \delta E^s}{1 - F'(E^s)} + \frac{((1 - \delta)(1 - F'(E^s)) - \alpha_1)}{1 - F'(E^s)} E_t$$

Proposition 2

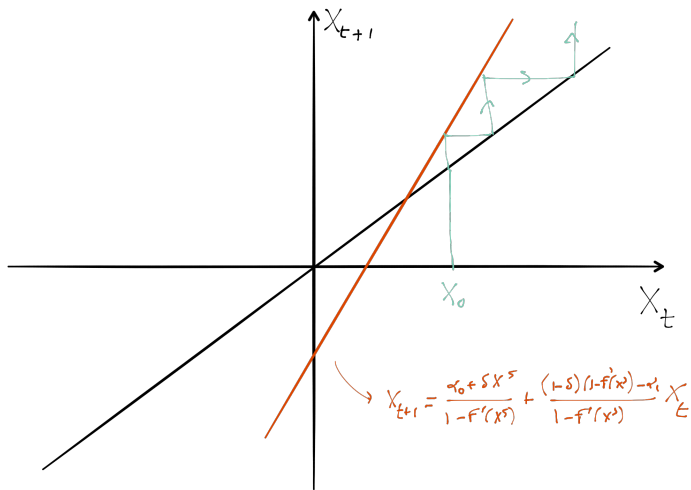
With strategic substitutability ($F'(\cdot) < 0$), the economy is always stable

Proposition 3

With strategic complementarity ($F'(\cdot) > 0$), there is always a level of complementarity smaller than 1 such that the SS is (locally) unstable.

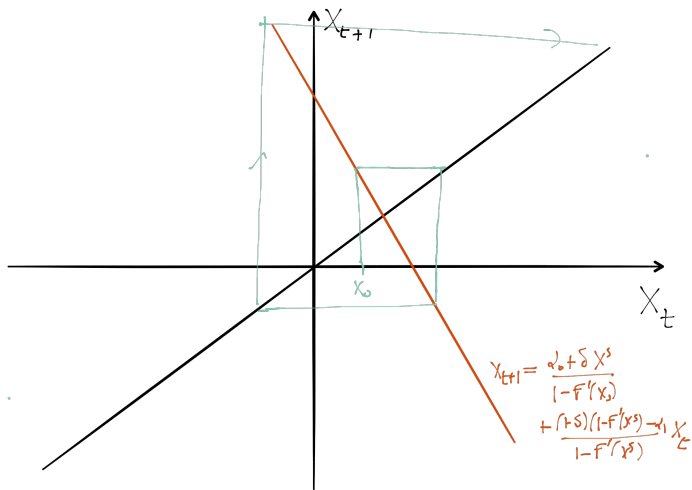
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One-Dimension Case



2. A framework for Dynamics with Strategic Interactions

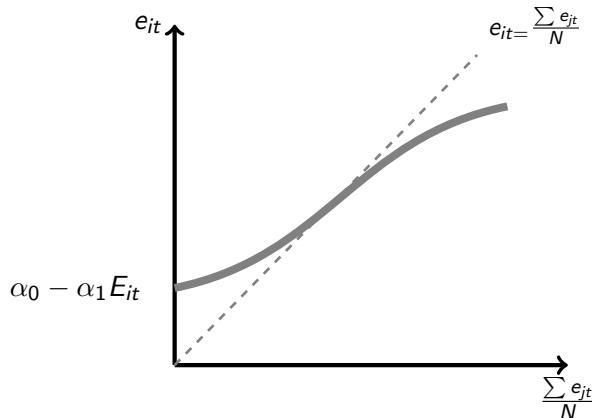
One-Dimension Case



2. A framework for Dynamics with Strategic Interactions

One-Dimension Case

- ▶ The SS E^s becomes locally unstable
- ▶ “Real data” does not look explosive
- ▶ Let's assume that strategic complementarities dies out when the economy is far above of below E^s
- ▶ F is “S-shaped”



2. A framework for Dynamics with Strategic Interactions

One-Dimension Case

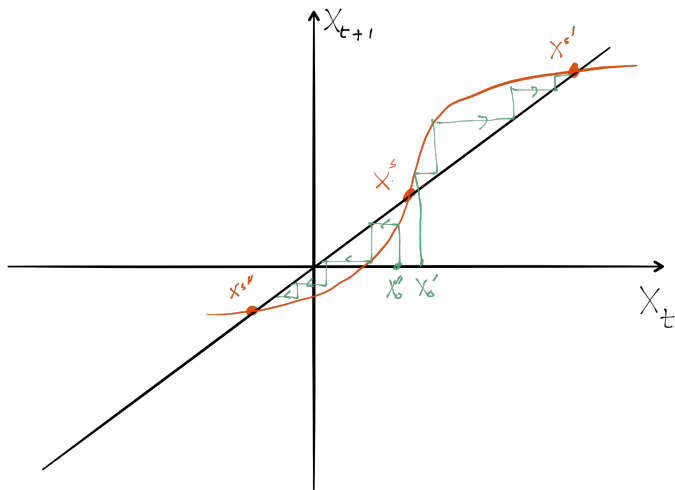
- ▶ Assume for simplicity that F is symmetric wrt E^s

$$E_{t+1} = \frac{\alpha_0 + \delta E^s}{1 - F'(E^s)} + \frac{((1 - \delta)(1 - F'(E^s)) - \alpha_1)}{1 - F'(E^s)} E_t$$

- ▶ The dynamics of the economy will very much depend on whether $\frac{((1 - \delta)(1 - F'(E^s)) - \alpha_1)}{1 - F'(E^s)}$ is positive or negative
 - × Positive : Hysteresis
 - × Negative : Limit cycle
- ▶ Need to think of environments that makes this slope is positive or negative

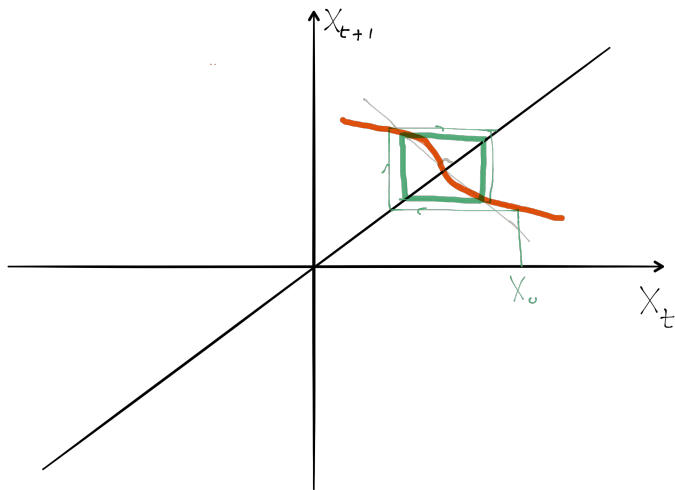
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One-Dimension Case



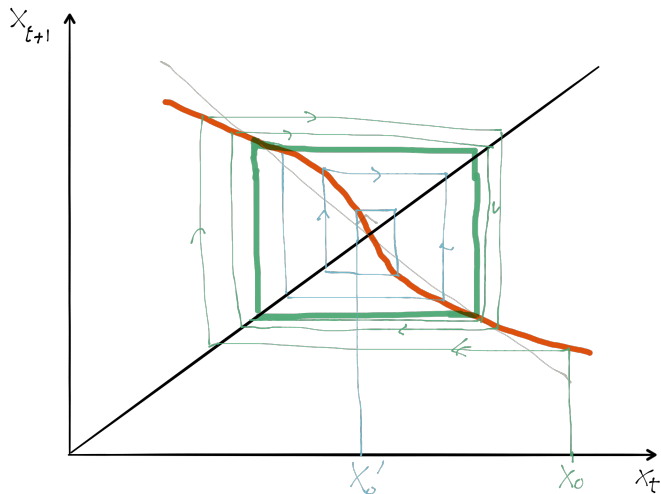
2. A framework for Dynamics with Strategic Interactions

One-Dimension Case



2. A framework for Dynamics with Strategic Interactions

One-Dimension Case



2. A framework for Dynamics with Strategic Interactions

One-Dimension Case

- ▶ Problem with one dimension model : the dynamics does not look like a business cycle
- ▶ E_t is *not* positively correlated.
- ▶ This needs not to be true in the more general model where $\alpha_2 > 0$

2. A framework for Dynamics with Strategic Interactions

Two-Dimension Case

$$\begin{aligned}E_{it+1} &= (1 - \delta)E_{it} + e_{it} \\e_{it} &= \alpha_0 - \alpha_1 E_{it} + \alpha_2 e_{it-1} + F(e_t)\end{aligned}$$

- ▶ Without strategic interactions ($F(\cdot) \equiv 0$)
- ▶ When all agent behave symmetrically :

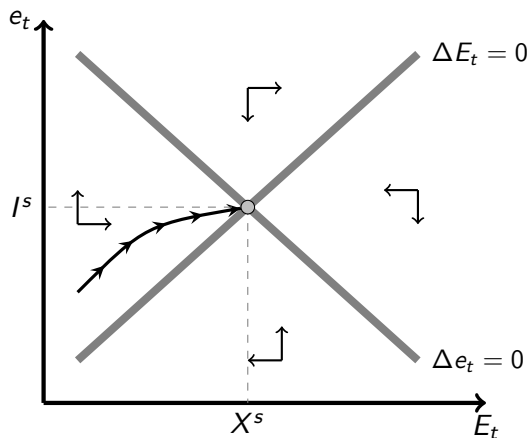
$$\begin{pmatrix} e_t \\ E_t \end{pmatrix} = \underbrace{\begin{pmatrix} \alpha_2 - \alpha_1 & -\alpha_1(1 - \delta) \\ 1 & 1 - \delta \end{pmatrix}}_{M_L} \begin{pmatrix} e_{t-1} \\ E_{t-1} \end{pmatrix} + \begin{pmatrix} \alpha_0 \\ 0 \end{pmatrix}$$

Proposition 4

Both eigenvalues of the matrix M_L lie within the unit circle. Therefore, the system is stable.

2. A framework for Dynamics with Strategic Interactions

Two-Dimension Case



Here we have assumed that the two eigenvalues are real and positive.

2. A framework for Dynamics with Strategic Interactions

Two-Dimension Case

- Math
- Dynamics is

$$\begin{pmatrix} e_t \\ E_t \end{pmatrix} = \underbrace{\begin{pmatrix} \alpha_2 - \alpha_1 & -\alpha_1(1-\delta) \\ 1 & 1-\delta \end{pmatrix}}_{M_L} \begin{pmatrix} e_{t-1} \\ E_{t-1} \end{pmatrix} + \begin{pmatrix} \alpha_0 + F(e_t) \\ 0 \end{pmatrix}$$

- Local dynamics is

$$\begin{pmatrix} e_t \\ E_t \end{pmatrix} = \underbrace{\begin{pmatrix} \frac{\alpha_2 - \alpha_1}{1 - F'(I^s)} & -\frac{\alpha_1(1-\delta)}{1 - F'(I^s)} \\ 1 & 1-\delta \end{pmatrix}}_M \begin{pmatrix} e_{t-1} \\ E_{t-1} \end{pmatrix} + \begin{pmatrix} \left(1 - \frac{\alpha_2 - \alpha_1}{1 - F'(I^s)}\right) I^s + \left(\frac{\alpha_1(1-\delta)}{1 - F'(I^s)}\right) X^s \\ 0 \end{pmatrix}$$

- When $F'(I^s)$ varies from $-\infty$ to 1, eigenvalues of M vary

2. A framework for Dynamics with Strategic Interactions

Two-Dimension Case

Proposition 5

As $F'(I^S)$ varies from 0 to $-\infty$, the eigenvalues of M always stay within the unit circle and therefore the system remains locally stable.

2. A framework for Dynamics with Strategic Interactions

Two-Dimension Case

Proposition 6

As $F'(I^s)$ varies from 0 towards 1, the dynamic system will become locally unstable.

(bifurcation)

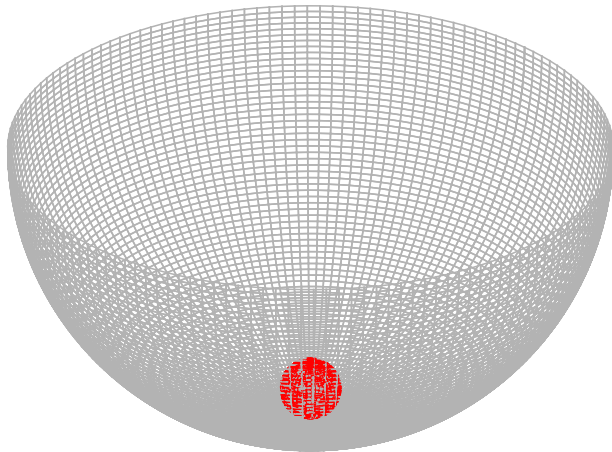
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Two-Dimension Case

- ▶ Here we can have a limit cycle with persistence
- ▶ Consider the steady state (E^s, I^s)
- ▶ Strategic complementarities : centrifugal force that pushes away from the steady state when close to.
- ▶ Accumulated variable X : centripetal force that pushes towards the steady state when away from.
- ▶ The steady state locally unstable, but forces push the economy back to the steady state when it is further from it.
- ▶ It is quite intuitive why a limit cycle occurs when the steady state moves from stable to unstable

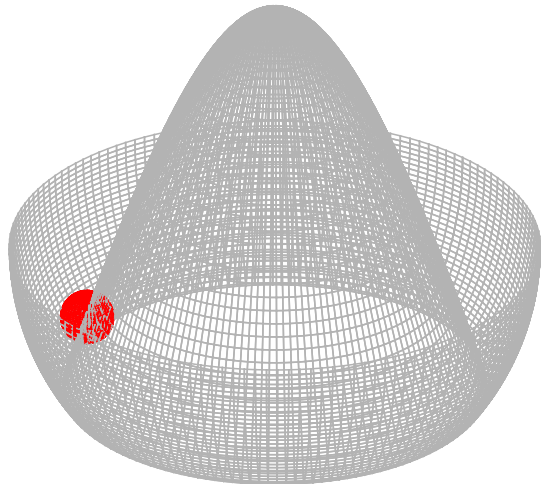
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FIGURE 1 – Stable Steady State



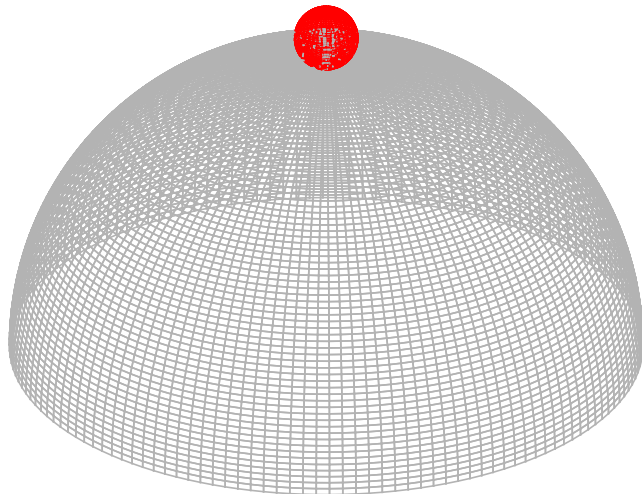
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FIGURE 2 – HOPF Supercritical bifurcation : Attractive Limit Cycle



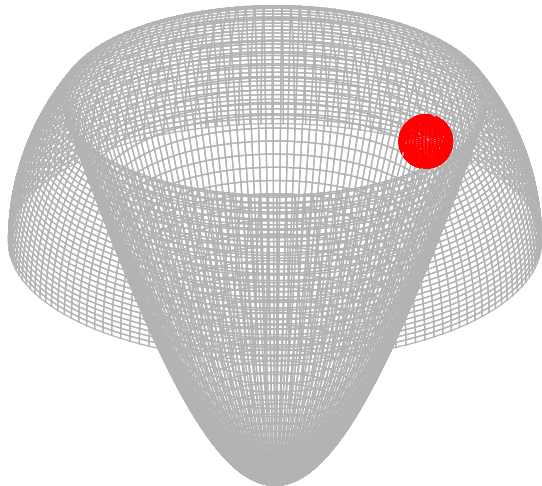
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FIGURE 3 – Unstable Steady State



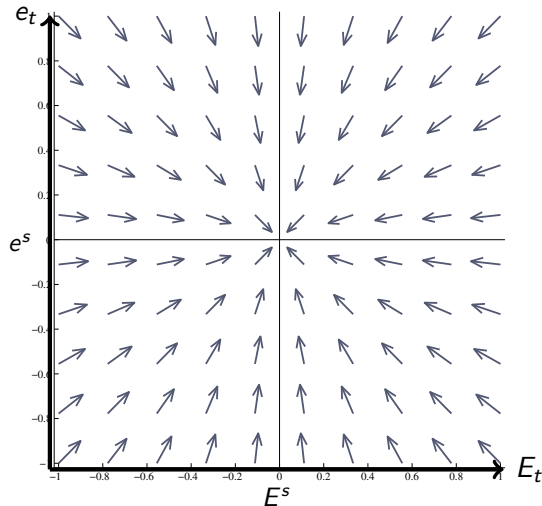
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FIGURE 4 – HOPF Subcritical bifurcation : Repulsive Limit Cycle



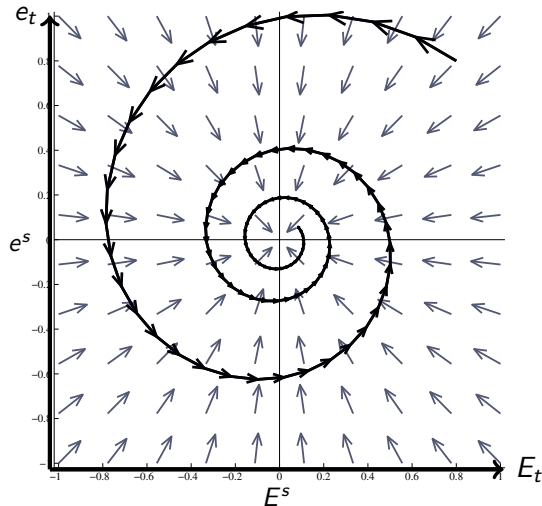
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Two-Dimension Case - Global stability



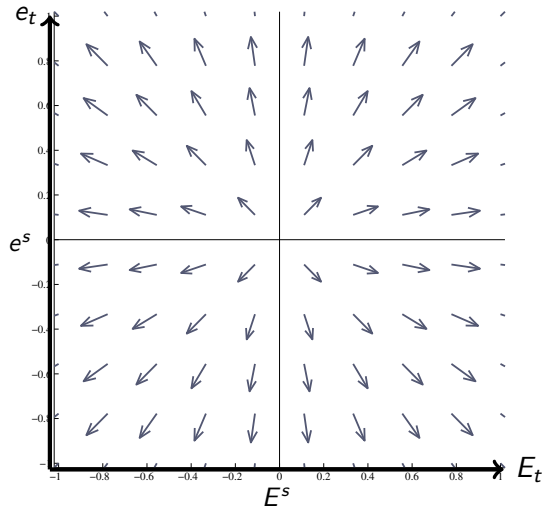
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Two-Dimension Case - Global stability



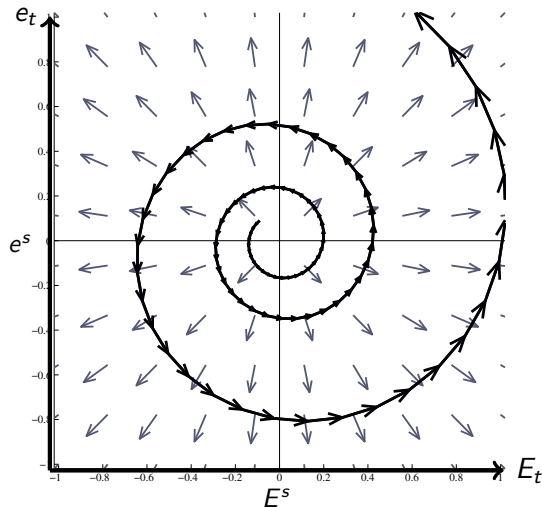
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Two-Dimension Case - Global instability



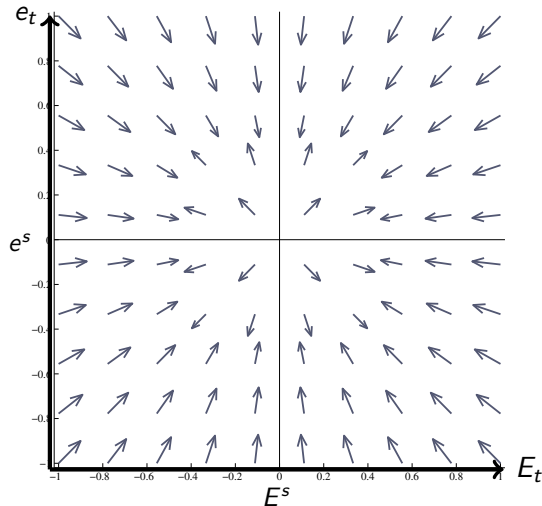
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Two-Dimension Case - Global instability



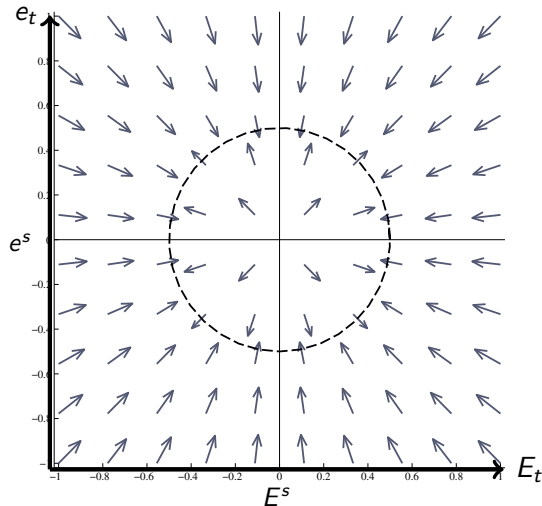
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Two-Dimension Case - Stable limit cycle



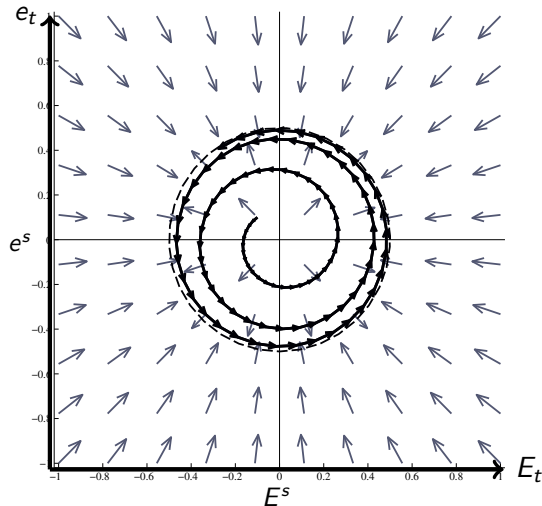
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Two-Dimension Case - Stable limit cycle



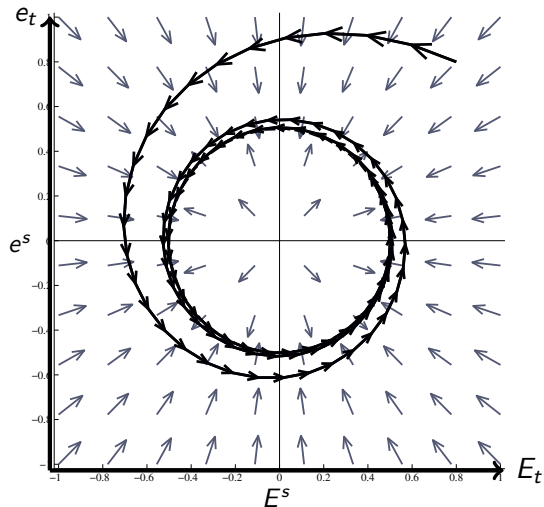
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Two-Dimension Case - Stable limit cycle



2. A framework for Dynamics with Strategic Interactions

Two-Dimension Case - Stable limit cycle



2. A framework for Dynamics with Strategic Interactions

Two-Dimension Case

- ▶ Discrete time version of the HOPF theorem.
- ▶ Consider the 2-dimensional smooth system

$$Y_{t+1} = F(Y_t; \alpha)$$

- ▶ Assume : Steady State for $\alpha = 0$: $F(\vec{0}; 0) = 0$
- ▶ Assume : Jacobian of F at $(0,0)$ has two eigenvalues $(1 + \alpha)e^{\pm i\theta_0}$
- ▶ That system has a bifurcation at $\alpha = 0$
- ▶ *i.e.* the Steady State changes stability with α
- ▶ Then, the system $Y_{t+1} = F(Y_t; \alpha)$ has a limit cycle near the origin.
- ▶ Note : other possibilities is that one eigenvalue is 1 or -1.

2. A framework for Dynamics with Strategic Interactions

Looking for the limit cycle

- ▶ Nice theorem : we simply have to look at the *linearized* dynamics to prove existence of a limit cycle
- ▶ The parameter α is here the degree of strategic complementarities at the steady state $F'(I^s)$
- ▶ It is quite intuitive why a limit cycle occurs when the steady state moves from stable to unstable

2. A framework for Dynamics with Strategic Interactions

Bifurcations

- ▶ 3 types of bifurcation
 - × Fold bifurcation : appearance of an eigenvalue equal to 1,
 - × Flip bifurcation : appearance of an eigenvalue equal to -1
 - × HOPF bifurcation : appearance of two complex conjugate eigenvalues of modulus 1.
- ▶ We are interested in HOPF bifurcation because the limit cycle will be “persistent”

2. A framework for Dynamics with Strategic Interactions

Bifurcations

Proposition 7

As $F'(I^s)$ varies from 0 towards 1, the dynamic system will become unstable and :

- ▶ *If $\alpha_2 > \alpha_1/(2 - \delta)^2$, then a HOPF (NEIMARK-SACKER) bifurcation will occur.*
- ▶ *If $\alpha_2 < \alpha_1/(2 - \delta)^2$, then a flip bifurcation will occur.*

2. A framework for Dynamics with Strategic Interactions

Bifurcations

- ▶ In the case of flip and HOPF bifurcation, a limit cycle appears
- ▶ In case of HOPF, the cycle is “smooth”
- ▶ Condition for an HOPF bifurcation :

$$e_{it} = \alpha_0 - \alpha_1 E_{it} + \alpha_2 e_{it-1} + F\left(\frac{\sum e_{jt}}{N}\right)$$

- × as δ approaches 0, $\alpha_2 > 1/4$
- × in general, what is needed is δ not too large and α_2 large enough

2. A framework for Dynamics with Strategic Interactions

Stability of the limit cycle

- In the case of the HOPF bifurcation, the limit cycle can be attractive (*the bifurcation is supercritical*) or repulsive (*the bifurcation is subcritical*)

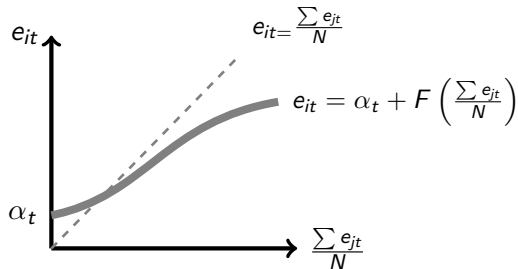
2. A framework for Dynamics with Strategic Interactions

Stability of the limit cycle

Proposition 8

If $F'''(I^s)$ is sufficiently negative, then the HOPF bifurcation will be supercritical. Therefore, the limit cycle is attractive.

- ▶
- ▶ $F''' < 0$ corresponds to an *S-shaped* reaction function



$$\alpha_t = \alpha_0 - \alpha_1 E_{it} + \alpha_2 e_{it-1}$$

3. Quantitative Reduced Form Limit Cycle Model

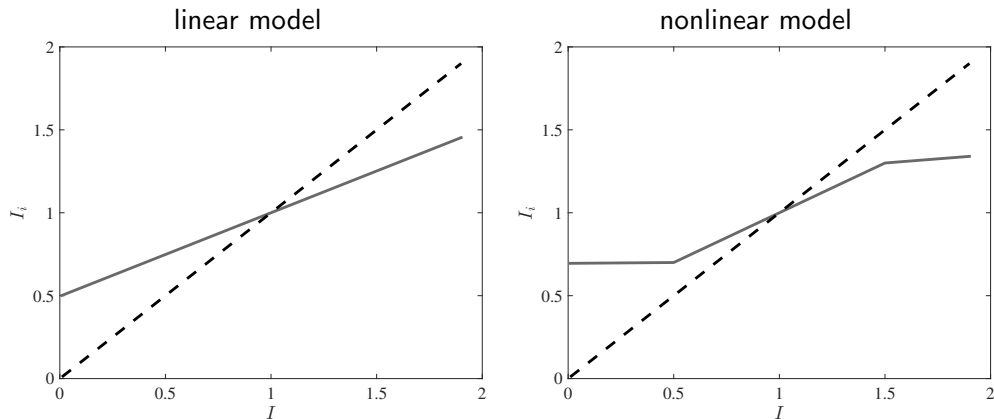
Model

- ▶ $E_{it+1} = (1 - \delta)E_{it} + e_{it}$
- ▶ $e_{it} = \alpha_0 - \alpha_1 E_{it} + \alpha_2 e_{it-1} + F\left(\frac{\sum e_{it}^s}{N}, u_t\right)$
- ▶ $F(e_t, u_t) = \tilde{F}(e_t) + u_t.$
- ▶ $u_t = \rho u_{t-1} + \varepsilon_t$
- ▶ $\tilde{F}$: S-shaped and piecewise linear

3. Quantitative Reduced Form Limit Cycle Model

Numerical example

FIGURE 5 – Best response rules in the numerical example



3. Quantitative Reduced Form Limit Cycle Model

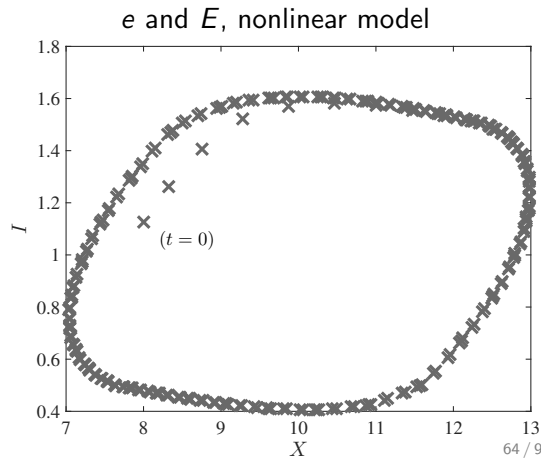
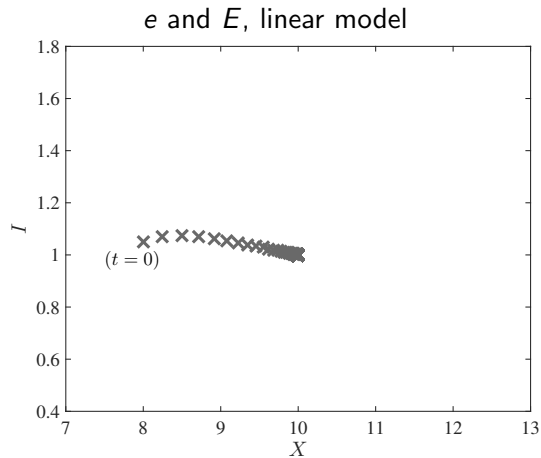
Numerical example

- ▶ Steady state is $e_{SS} = 1$, $E_{SS} = 10$
- ▶ Deterministic simulation : let $E_0 = 8$, $e_0 = 1$

3. Quantitative Reduced Form Limit Cycle Model

Numerical example

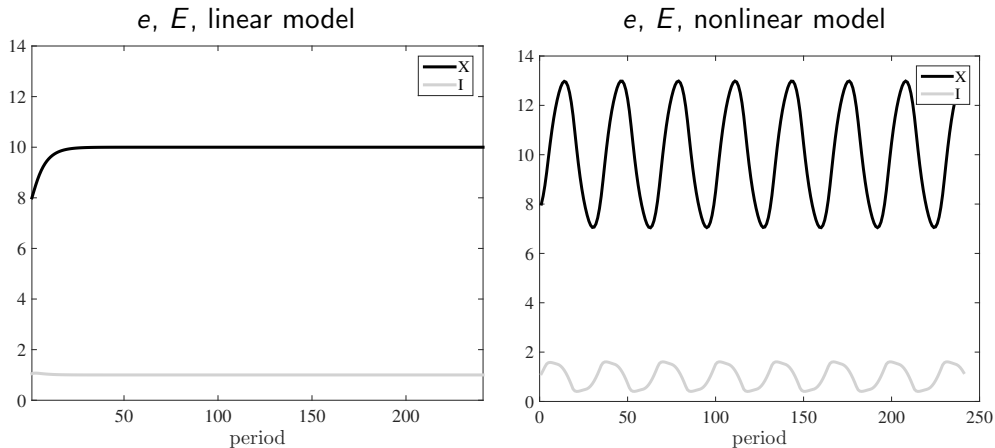
FIGURE 6 – Deterministic simulation



3. Quantitative Reduced Form Limit Cycle Model

Numerical example

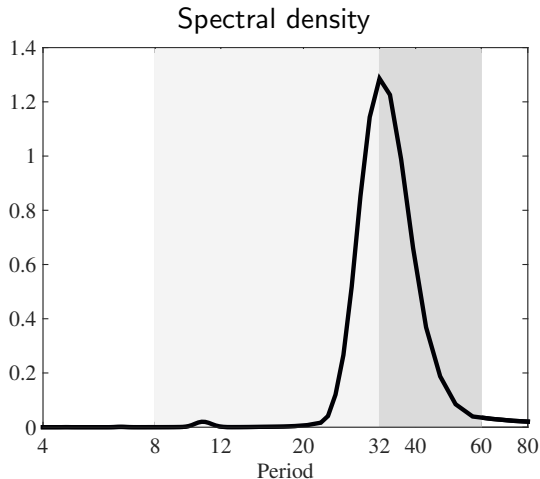
FIGURE 7 – Deterministic simulation



3. Quantitative Reduced Form Limit Cycle Model

Numerical example

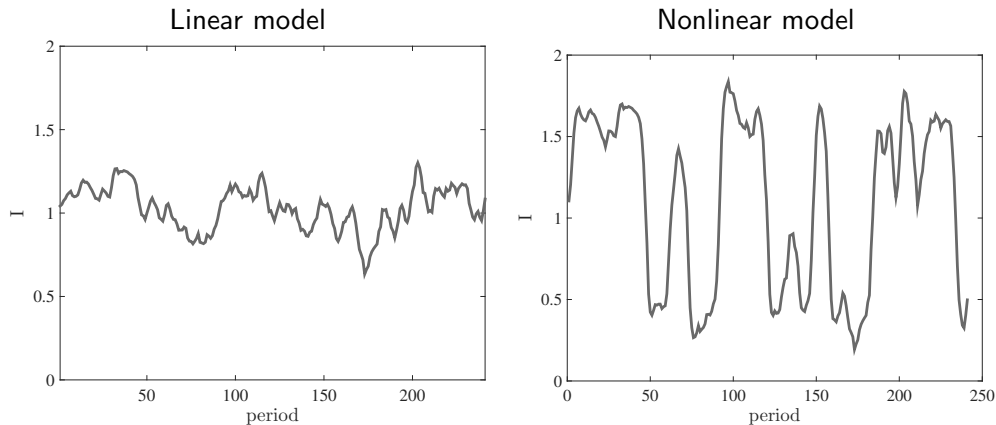
FIGURE 8 – Deterministic simulation of the nonlinear model



3. Quantitative Reduced Form Limit Cycle Model

Numerical example

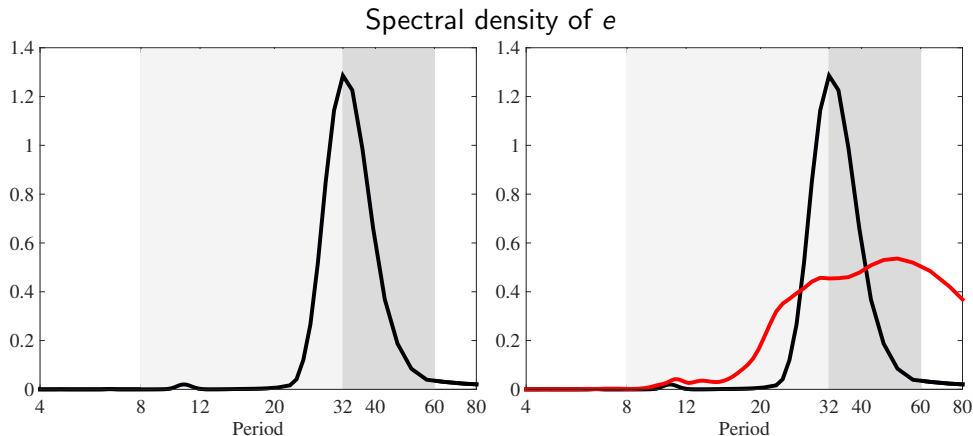
FIGURE 9 – One stochastic simulation



3. Quantitative Reduced Form Limit Cycle Model

Numerical example

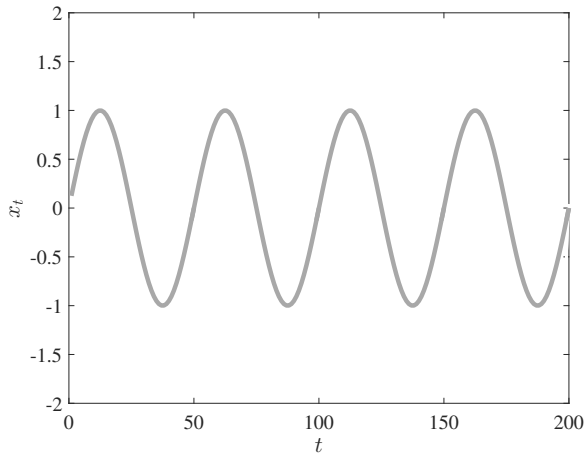
FIGURE 10 – Deterministic and Stochastic simulation of the nonlinear model



3. Quantitative Reduced Form Limit Cycle Model

What the results are not

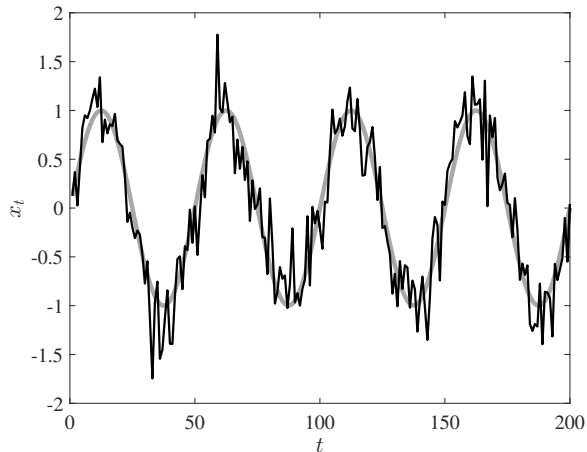
FIGURE 11 – $x_t = \sin(\omega t)$



3. Quantitative Reduced Form Limit Cycle Model

What the results are not

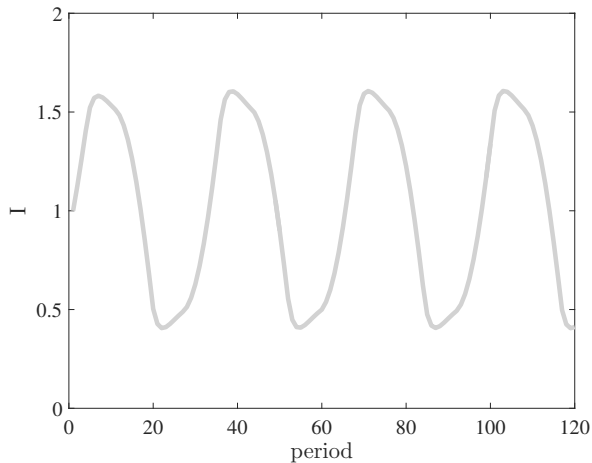
FIGURE 12 – $x_t = \sin(\omega t) + u_t$



3. Quantitative Reduced Form Limit Cycle Model

What the results are

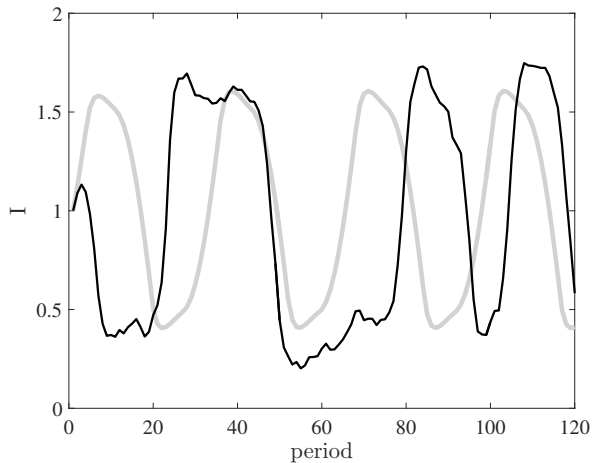
FIGURE 13 – Reduced Form Model



3. Quantitative Reduced Form Limit Cycle Model

What the results are

FIGURE 14 – Reduced Form Model



4. Adding forward looking elements



$$e_{it} = \alpha_0 - \alpha_1 E_{it} + \alpha_2 e_{it-1} + \alpha_3 E_t[e_{it+1}] + F\left(\frac{\sum e_{jt}}{N}\right)$$

- ▶ with accumulation remaining the same

$$E_{it} = (1 - \delta)E_{it} + e_{it}$$

- ▶ Restrict attention to situations where this system is saddle path stable absent of complementarities.
- ▶ The local dynamics is described by the 3 eigenvalues of the linearized system

4. Adding forward looking elements

Set of results when one increases ρ

Proposition 9

If unique steady state, then no indeterminacy nor Fold bifurcations (always one positive and greater than one eigenvalue)

4. Adding forward looking elements

Set of results when one increases ρ

Proposition 10

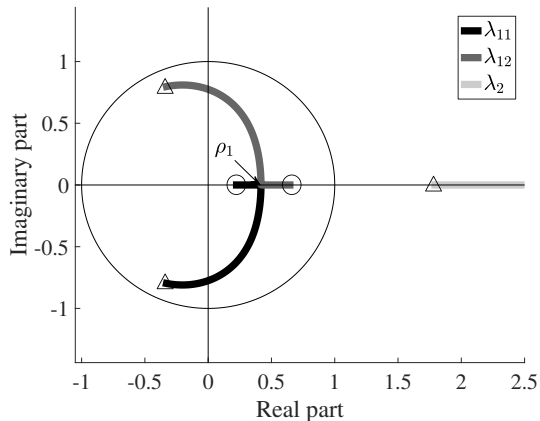
If α_2 (sluggishness) sufficiently large, then the two other eigenvalues will become complex and

- ▶ *either will stay inside the unit disk (non locally explosive cycles)*
- ▶ *or will exit the unit disk (HOPF bifurcation and limit cycles)*

4. Adding forward looking elements

Set of results when one increases ρ

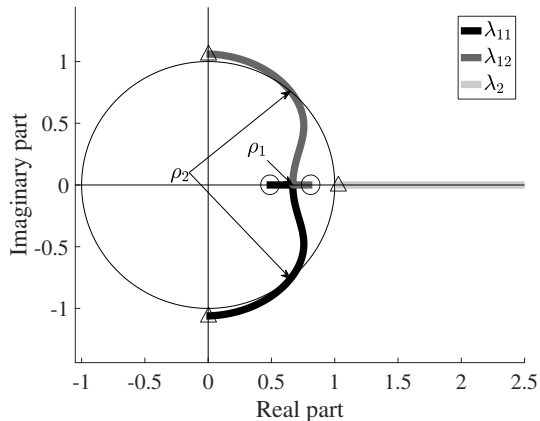
FIGURE 15 – Increasing complementarities ρ : local stability



4. Adding forward looking elements

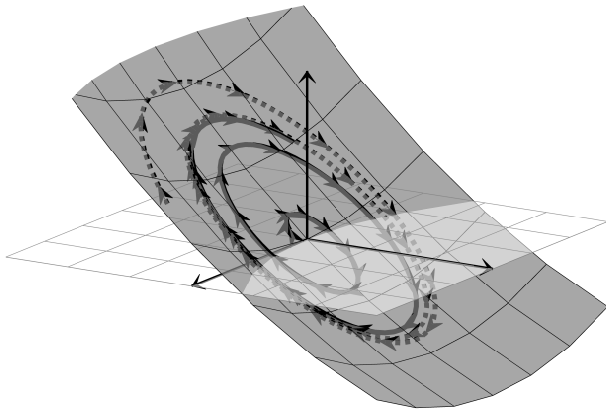
Set of results when one increases ρ

FIGURE 16 – Increasing complementarities ρ : local instability and limit cycle



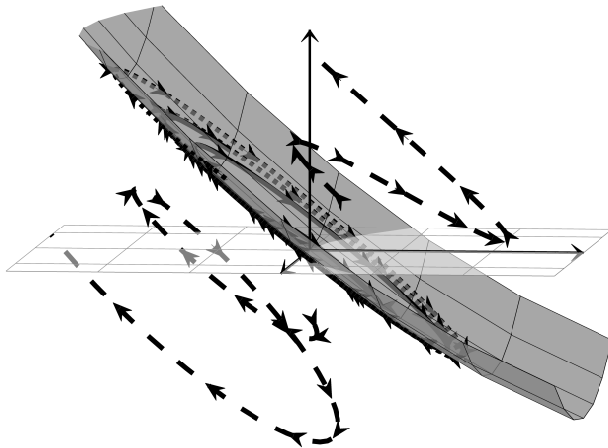
4. Adding forward looking elements

FIGURE 17 – A Saddle Limit Cycle



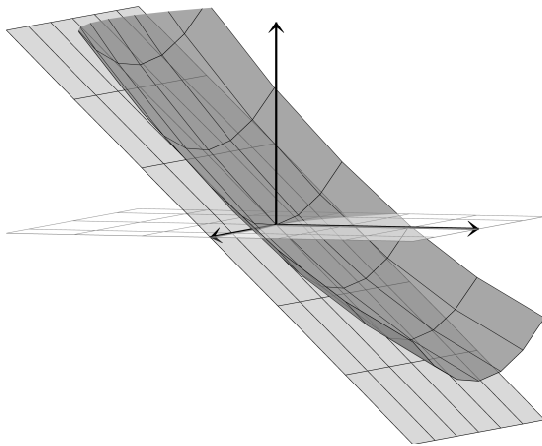
4. Adding forward looking elements

FIGURE 18 – Explosive path



4. Adding forward looking elements

FIGURE 19 – Linear and non linear stable manifolds



4. Adding forward looking elements

- ▶ Weak complementarities + dynamics create cycles
- ▶ It can also create local instability and limit cycles
- ▶ One should be aware that estimating linear models is likely to bias against this mechanism as one rules out local instability.

5. A NK Model with Financial Frictions

$$\begin{cases} e_{it} &= \alpha_0 - \alpha_1 E_{it} + \alpha_2 e_{it-1} + \alpha_3 E_t[a_{it+1}] + F\left(\frac{\sum e_{jt}}{N}\right) \\ E_{it} &= (1 - \delta)E_{it} + e_{it} \end{cases}$$

- ▶ Stylized NK model which is extended to allow for the forces highlighted in our general structure.
- ▶ We add accumulation of durable-housing goods and habit persistence : *accumulation and sluggishness*
- ▶ Financial frictions : households face unemployment risk and cannot fully insure against it $\rightsquigarrow$ counter-cyclical risk premium $\rightsquigarrow$ *complementarities*
- ▶ Estimate parameters based on spectrum observations and higher moments. (use perturbation method and indirect inference)
- ▶ AR(1) discount factor shock, with persistence ρ

5. A NK Model with Financial Frictions

Allowing for durable goods and accumulation

- ▶ The representative household who can buy/rent consumption services from the market.
- ▶ Households consumption services are supplied by firms and come from either the production of new goods or from the service flow of existing durables.

$$C_t = sX_t + F(L_t\theta_t)$$

where X_t is the stock of durable goods and $F(l_t\theta_t)$ is the current production of goods. The accumulation of durables is

$$X_{t+1} = (1 - \delta)X_t + \psi F(L_t\theta_t)$$

where a fraction $\psi \leq (1 - \delta)$ of newly produced goods become durables after one period.

- ▶ Households own the stock of durable-housing, rent it to firms, who supply back the consumption services as a composite good.

5. A NK Model with Financial Frictions

Risk premium & Financial Frictions

- ▶ Labor indivisibility $\rightsquigarrow$ some workers are unemployed
- ▶ Spending decisions are made before labor market outcome is known
- ▶ The large family pays only for a fraction of its unemployed members.
- ▶ The interest rate which household face is then equal to the policy rate plus a risk premium

$$(1 + r_t) = (1 + i_t + r_t^p)$$

- ▶ The risk premium is a decreasing function of the economic activity, where the output gap and employment gap are interchangeable

$$r_t^p = G(\ell_t)$$

- ▶ The interest premium is generically a non-linear function of activity.
- ▶ The households Euler will have the familiar form (assuming external habit)

$$U'(C_t - \gamma C_{t-1}) = \beta_t E_t U(C_{t+1} - \gamma C_t)(1 + r_t)$$

5. A NK Model with Financial Frictions

NK Model Framework

- ▶ Firms face stochastic opportunities to change price (CALVO)
- ▶ Have Taylor rule of form

$$i_t = E_t \pi_{t+1} + \phi E_t \ell_{t+1}$$

where ℓ_t is (log) employment or output gap, and $U(\cdot) = \ln(\cdot)$

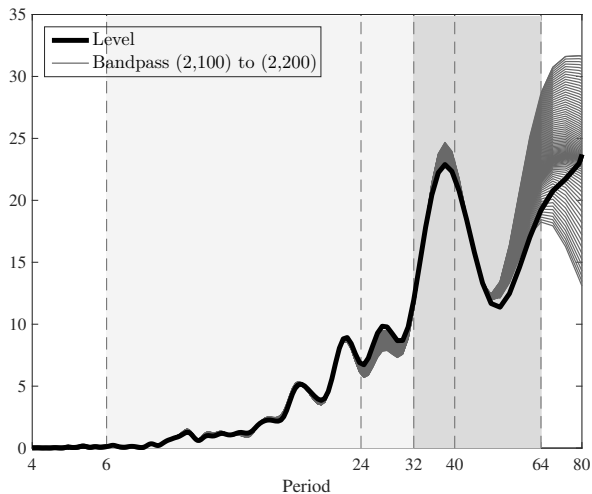
6. Results From Estimation

Estimation

- ▶ Estimate parameters of model by SMM
- ▶ Targets
 - × spectrum of hours worked on the frequencies 2-50
 - × spectrum of interest rate spread on the frequencies 2-50
 - × Set of other higher moments (correlation, kurtosis and skewness of hours and spread)
- ▶ We estimate the model
 - × Without complementarities (No Risk Premium)
 - × With complementarities, but linear approx.
 - × With complementarities, but order 3 approx.

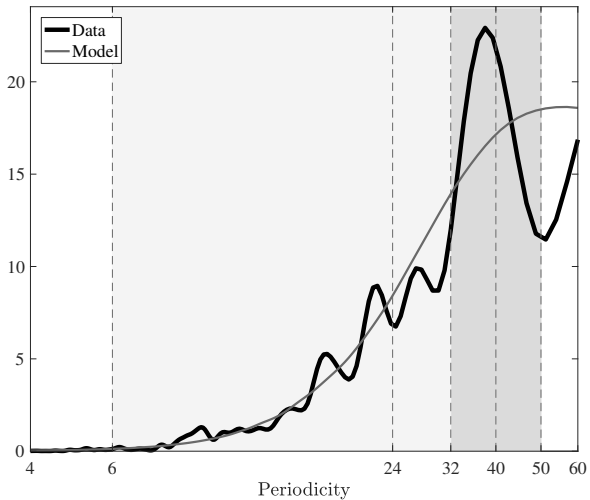
6. Results From Estimation

FIGURE 20 – Spectral Density, Level and Bandpass (2,100) to (2,200), Full Sample - Non Farm Business Hours



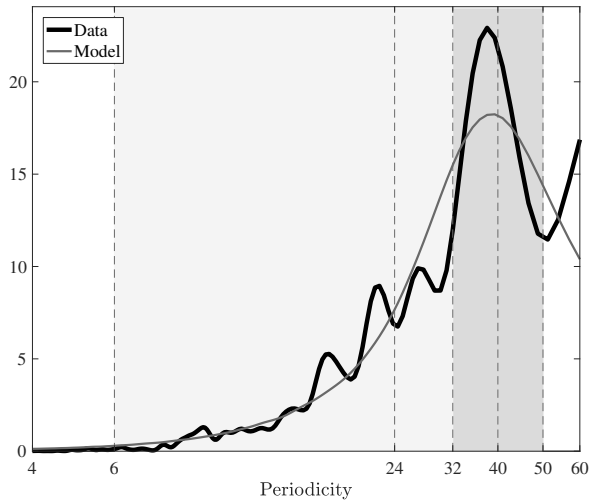
6. Results From Estimation

FIGURE 21 – Spectrum of Hours, No Risk Premium



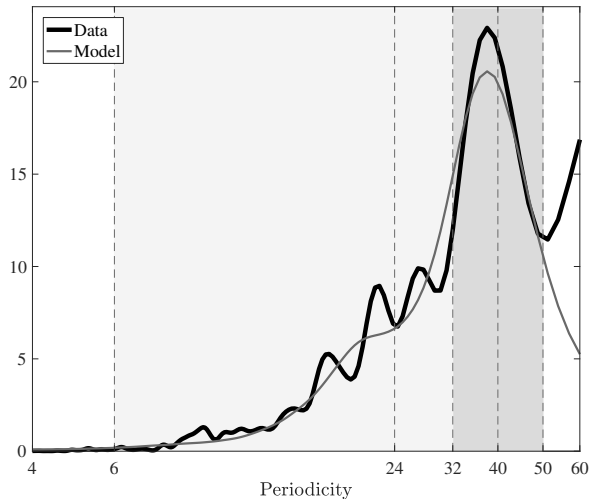
6. Results From Estimation

FIGURE 22 – Spectrum of Hours, Linear Risk Premium



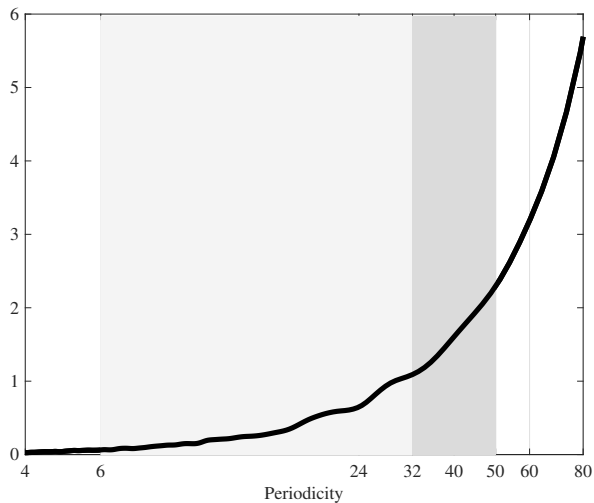
6. Results From Estimation

FIGURE 23 – Spectrum of Hours, Non-Lin. Risk Premium



6. Results From Estimation

FIGURE 24 – Hours Spectrum in Smets & Wouters' Model



6. Results From Estimation

TABLE 1 – Parameter Estimates

		Non-Lin. RP	Lin. RP	No RP
ω	CRRA	0.2997	0.2408	0.2408*
γ	Habit	0.5335	0.5876	0.8346
ψ	Initial deprec.	0.4000	0.2994	0.1154
ϕ	Taylor rule	0.0421	0.0467	0.0000
R_2	Risk prem.	0.00017	—	—
R_3	Risk prem.	0.00006	—	—
$\bar{f}$	Debt backing	0.8668	0.8827	1*
b	Default cost	0.2870	0.3568	—
ρ	Autocorr.	-0.0000	0.1387	0.8346
σ	Innov. s.d.	0.0157	0.0272	0.0335
$s.d.(\mu)$	Uncond. s.d.	0.0157	0.0275	0.0609

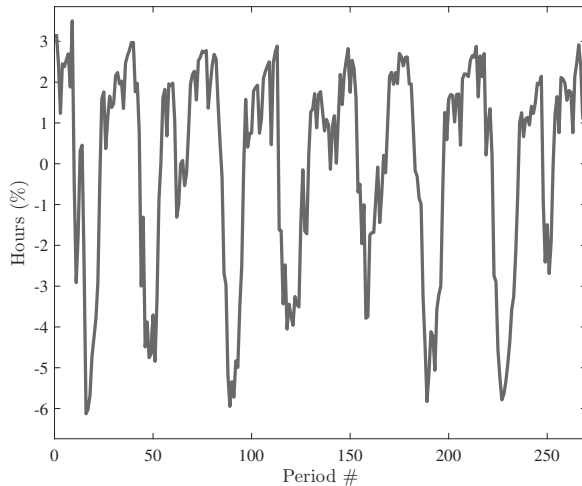
6. Results From Estimation

TABLE 2 – Eigenvalues

	Non-Lin. RP	Lin. RP	No RP
λ_1, λ_2	$1.1032 \pm 0.2164i$	$0.9258 \pm 0.1372i$	0.8346,
$ \lambda_j $	1.1242	0.9359	0.8346

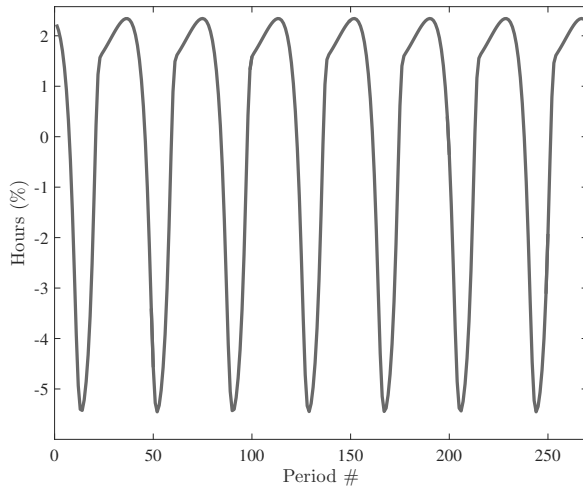
6. Results From Estimation

FIGURE 25 – Sample Draw for Hours



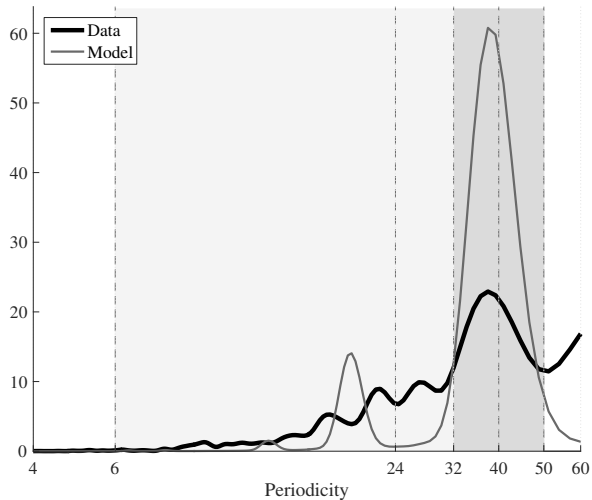
6. Results From Estimation

FIGURE 26 – Sample Draw for Hours, no shocks



6. Results From Estimation

FIGURE 27 – Spectrum for Hours, no shocks



6. Results From Estimation

- ▶ Shocks are important for explaining the data.
- ▶ However, we estimate them to be i.i.d. in case with limit cycle.
 - × Almost all dynamics are internal.
 - × Main purpose : makes cycle *irregular*.
- ▶ Important : Adding shocks to a model with a limit cycle does not simply result in a predictable cycle with noise around it.
 - × Shocks have temporary amplitude effect, but “random-walk” effect on phase.
 - × i.e., permanently shift the whole cycle forward or backward in time.
 - × Phase of cycle completely unpredictable in the infinite future.

6. Results From Estimation

Estimating over different frequency windows

- ▶ Estimate the linear and non linear version over alternative frequencies : 2-32, 2-50, 2-60, 2-100.
- ▶ Main results
 - × over 2-32 : come back to case where the exogenous driving forces is the cycle
 - × Our results not changed much if go to 2-60 or 6-50, 6-60.. Almost all dynamics are internal
 - × However, if extend further, example 2-100, back to exogenous driving force dominating

6. Results From Estimation

Estimating over different frequency windows

FIGURE 28 – Estimation over 2-100

